



Descriptive Statistics

Understanding and summarizing data clearly

Simple explanations • worked examples • practical use cases



1. What descriptive statistics does

Descriptive statistics converts raw data into a clear summary. It answers questions such as: What is typical? How widely do values vary? Where does a value sit within the dataset? What shape does the distribution have?

It describes the data you already have. It does not, by itself, prove causes or make conclusions about a larger population.

Four useful groups

Group	Measures	Purpose
Central tendency	Mean, median, mode	Find a typical value
Spread	Range, variance, standard deviation, IQR	Measure variability
Position	Quartiles, percentiles	Locate a value
Shape	Skewness, distribution	Understand balance and unusual tails

2. Measures of centre

Worked example: daily branch transactions

Values: 80, 90, 100, 110, 520.

$$\text{Mean} = (80 + 90 + 100 + 110 + 520) / 5 = 180$$

The median is 100 because it is the middle ordered value. The large value 520 pulls the mean upward, so the median better represents a typical day here.

- Mean uses every value but is sensitive to extreme values.
- Median is resistant to extreme values.
- Mode is the most frequently occurring value and also works with categories.

3. Measures of spread and position

For transaction amounts 10, 20, 30, 40 and 50, the mean is 30.

Population variance = 200; standard deviation = $\sqrt{200} = 14.14$

The standard deviation is expressed in the original unit, so values typically lie about INR 14.14 from the mean. The range is $50 - 10 = 40$.

Quartiles

- Q1: 25% of values are at or below this point.
- Q2: the median; 50% of values are at or below it.
- Q3: 75% of values are at or below it.
- IQR = $Q3 - Q1$; useful for identifying unusual values.

4. Banking use case and interpretation

Use case: ATM cash-withdrawal monitoring

A bank summarizes daily withdrawals per ATM using count, mean, median, standard deviation, quartiles, minimum and maximum. A large gap between mean and median can signal skewness; a high standard deviation can indicate volatile cash demand.

Finding	Possible interpretation	Operational action
Mean much higher than median	A few very large withdrawals	Review limits and cash planning
High standard deviation	Demand changes sharply by day	Increase safety stock or forecast by day
Value above $Q3 + 1.5 \times IQR$	Potential outlier	Validate event, holiday or anomaly
Stable median and IQR	Predictable typical demand	Use standard replenishment plan

Do not summarize a dataset with only one number. Centre, spread, position and shape together provide a more reliable picture.